ModelIt!

Teacher's Guide

Earth Has a Fever

Why 2 Degrees Could Change Everything

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-ESS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-5

Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask questions to clarify evidence about factors causing global temperature rises.
Disciplinary Core Idea	**ESS3.D Global Climate Change** — Human activities have significantly altered the biosphere, sometimes damaging or destroying natural habitats and causing extinction of species.
Crosscutting Concept	**Stability and Change** — Students analyze how small changes in CO_2 concentration create large, cascading changes in global climate systems.

CAST Testing Alignment

MULTIPLE CHOICE:

Since 1850, atmospheric CO_2 has increased from 280 ppm to over 420 ppm. During the same period, average global temperature has risen about 1.1°C. Which statement best explains this relationship?

CONSTRUCTED RESPONSE:

Using evidence from your climate model, explain why scientists are concerned about a 2°C rise in global temperature. Include at least one feedback loop in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Climate change is one of the defining challenges of the 21st century. Understanding the science behind it — greenhouse gases, feedback loops, and cascading effects — is essential for making informed decisions about our planet's future.

The Greenhouse Effect

Certain gases in Earth's atmosphere (CO_2, methane, water vapor) allow sunlight to pass through but trap heat radiating back from Earth's surface. This natural greenhouse effect keeps Earth habitable — without it, average temperature would be -18°C. The problem is that human activities are dramatically increasing greenhouse gas concentrations.

CO_2 and Temperature

Atmospheric CO_2 has increased from 280 ppm (pre-industrial) to over 420 ppm today — a 50% increase. Ice core data shows that CO_2 and temperature have moved together for 800,000 years. The current rate of CO_2 increase is unprecedented in the geological record.

Feedback Loops

Climate change involves powerful feedback loops. As temperatures rise, ice melts, exposing dark ocean and land that absorb MORE heat (ice-albedo feedback). Warming permafrost releases stored methane, causing MORE warming. These positive feedbacks can accelerate change beyond what CO_2 alone would cause.

Impacts of 2°C Warming

At 2°C above pre-industrial levels: sea levels could rise 0.5m+ (flooding coastal cities), extreme weather events become more frequent, coral reefs face mass extinction, agricultural zones shift, and water scarcity affects billions. The difference between 1.5°C and 2°C is enormous in terms of human impact.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify CO_2 concentration, solar energy, global temperature, and ice sheet coverage as key components of the climate system.

E **Establish Relationships**
Students determine that CO_2 positively affects temperature, which negatively affects ice coverage, creating a cascade of effects.

V **Visualize & Model**
Students build a climate model showing how greenhouse gas increases drive temperature and ice changes.

E **Evaluate Outcomes**
Students run pre-industrial, current trajectory, and emission reduction scenarios to compare outcomes.

R **Revise & Extend**
Students add methane emissions or albedo effect to explore feedback loops in the climate system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Earth has a fever. And we gave it to her. Let's understand why."*

Watch For

Show before/after photos of glaciers from the same location decades apart.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the biggest science story of your lifetime."*

Watch For

Have students read objectives. Pre-teach 'greenhouse effect' and 'feedback loop.'

Slide 3: Big Question

3 min

Say: *"Your body temperature varies by 1-2 degrees daily. No big deal. But for Earth? It changes EVERYTHING."*

Watch For

Show graph of global temperature over last century. Discuss scale.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a climate model to see why small changes have huge effects."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What controls Earth's temperature? What can humans influence?"*

Watch For

Guide sorting. Discuss why CO_2 is external (human-controlled input).

Slide 6: Activity 2: Connections

8 min

Say: *"More CO_2 means more heat trapped. But what does that DO to ice?"*

Watch For

Help students trace the cascade from CO_2 to temperature to ice.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare pre-industrial Earth to today. What does our model predict?"*

Watch For

Students predict first. Compare scenarios side by side.

Slide 8: Discoveries

5 min

Say: *"Why do scientists say climate change is accelerating? What did our model reveal?"*

Watch For

Lead discussion about feedback loops. Connect to real data.

Slide 9: STEM Challenge

12 min

Say: *"You have the science. Now design a solution for YOUR school."*

Watch For

Groups design climate action plans with specific targets.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** CO2 Concentration, Solar Energy Input
- **INTERNAL:** Average Global Temperature, Ice Sheet Coverage

Relationships

- **CO2 Concentration to Average Global Temperature:** POSITIVE (+) — More CO2 in the atmosphere traps more heat through the greenhouse effect, causing average temperatures to rise.
- **Solar Energy Input to Average Global Temperature:** POSITIVE (+) — More solar energy hitting Earth's surface increases the amount of heat available, raising temperatures.
- **Average Global Temperature to Ice Sheet Coverage:** NEGATIVE (-) — As global temperatures rise, ice sheets and glaciers melt, reducing the total area of Earth covered by ice.

Key Discoveries

- CO2 acts like a blanket — more CO2 traps more heat, raising global temperature
- Rising temperatures melt ice sheets, which reduces Earth's reflectivity and causes MORE warming (positive feedback)
- Even 2°C of warming dramatically changes weather patterns, sea levels, and ecosystems
- Human activities have increased atmospheric CO2 by over 50% since the Industrial Revolution

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Weather and climate are the same thing"	Weather is short-term, local conditions (today's temperature). Climate is long-term, global patterns (average temperature over decades). A cold day doesn't disprove climate change — it's the long-term TREND that matters.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The greenhouse effect is bad"	The NATURAL greenhouse effect is essential — without it, Earth would be frozen. The problem is that humans have ENHANCED it by adding excess CO2, trapping more heat than the system can handle.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Scientists disagree about climate change"	Over 97% of climate scientists agree that human activities are causing global warming. The scientific consensus is overwhelming. Disagreement exists about exact timelines and solutions, not about whether it's happening.